

The EU Carbon Border Adjustment Mechanism and China's Strategic Response: Carbon Pricing Realignment and Industrial Decarbonization

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Abstract—The European Union's Carbon Border Adjustment Mechanism (CBAM), entering its definitive phase in January 2026, operationalizes Nordhaus's climate club logic by linking market access to carbon pricing. This paper examines how CBAM reshapes China's carbon pricing architecture and sectoral decarbonization in steel and cement, the two sectors driving China's CBAM exposure. Using plant-level data from the Global Energy Monitor trackers (458 steel and 1,016 cement plants) and a liability model derived from Regulation EU 2023/956, we identify three transmission channels: a direct cost channel, a revenue retention incentive, and a signalling channel benchmarking CN-ETS ambition against EU ETS prices. At the prevailing price gap, BF-BOF steel exports face a net liability of EUR 140.7/t, versus EUR 6.7/t for EAF steel and EUR 42.1/t for cement—a 21:1 per-tonne liability ratio between BF-BOF and EAF steel that maps directly onto competitiveness exposure. CN-ETS reforms announced since 2023—absolute caps by 2027, sectoral expansion, and gradual auctioning—correlate with, and are plausibly reinforced by, CBAM's fiscal logic, though domestic dual-carbon goals provide independent momentum and causal attribution lies beyond this study's scope. We contribute an empirically calibrated assessment of CBAM's first-year implementation and identify conditions under which trade-linked enforcement complements rather than substitutes for multilateral cooperation.

Keywords: CBAM; CN-ETS; carbon pricing convergence; industrial decarbonization

Abbreviations: BF-BOF: Blast Furnace–Basic Oxygen Furnace; CBAM: Carbon Border Adjustment Mechanism; CCER: China Certified Emission Reductions; CCS/CCUS: Carbon Capture and Storage / Utilization and Storage; CGE: Computable General Equilibrium; CN-ETS: China Emissions Trading System; EAF: Electric Arc Furnace; EF: Emission Factor; EU ETS: European Union Emissions Trading System; GEM: Global Energy Monitor; GMRIO: Global Multi-Regional Input–Output; IAM: Integrated Assessment Model; MEE: Ministry of Ecology and Environment (China); MRV: Monitoring, Reporting, and Verification; Mtpa: Million tonnes per annum; NDRC: National Development and Reform Commission (China); NDC: Nationally Determined Contribution; SCM: Supplementary Cementitious Material; SOE: State-Owned Enterprise.

I. INTRODUCTION

A. From Free-Riding to Trade-Linked Enforcement: Why CBAM Matters

Global climate governance has long suffered from the tragedy of the atmospheric commons: mitigation benefits are universally shared, while abatement costs fall exclusively on individual jurisdictions (Hardin, 1968). This structural free-riding incentive has undermined every multilateral architecture, from the Kyoto Protocol's unenforceable targets to the Paris Agreement's voluntary NDCs. Nordhaus's (2015)

climate club proposal addresses this enforcement gap by proposing a coalition of high-ambition jurisdictions that adopt a common internal carbon price and impose uniform border tariffs on non-members, making free-riding costlier than abatement.

For nearly a decade, the climate club remained a theoretical construct. This changed with the CBAM, which entered its transitional phase in October 2023 and definitive phase in January 2026—the first instrument to operationalize trade-linked carbon enforcement at scale (EU, 2023). Under Regulation EU 2023/956, importers of six carbon-intensive goods must surrender certificates priced at the weekly average EU ETS allowance price. Critically, verified carbon costs paid in the country of origin are deducted euro for euro from CBAM liability. This deduction clause is CBAM's strategic core: it transforms cross-border price differentials into a direct fiscal incentive for non-EU jurisdictions to strengthen domestic carbon pricing and retain revenue locally.

B. CBAM's Particular Relevance for China

As the EU's largest non-EU source of CBAM-liable imports, China represents the most consequential test case for the climate club model. Three factors define this relationship.

First, the carbon price gap is large and policy-actionable. In 2024–2025, EU ETS allowances traded at EUR 50–75/tCO₂ (EU, 2023), while CN-ETS allowances cleared at EUR 10–13/tCO₂ (MEE, 2023). At an EU ETS price of EUR 80/tCO₂, steel produced via the blast furnace–basic oxygen furnace (BF-BOF) route—which converts iron ore into steel using coking coal as the primary reductant, generating approximately 2.10 tCO₂ per tonne of steel and offering limited near-term decarbonization options short of full technology replacement or carbon capture—faces a net liability of EUR 140.7/t. By contrast, the electric arc furnace (EAF) route—which melts recycled scrap steel using electricity, emitting only 0.10 tCO₂/t (Scope 1), and capable of near-zero emissions on a low-carbon grid, though constrained by scrap availability and grid carbon intensity—faces just EUR 6.7/t: a 21:1 per-tonne liability ratio that directly maps technology choice to competitiveness exposure.

Second, China's industrial structure creates structural vulnerability. Its steel sector (458 plants, 1,005 Mtpa capacity, 54% of global total) is 83.1% BF-BOF based, compared to 44% in the EU and 29% globally (WorldSteel, 2024). Its cement industry (1,016 plants, 1,772 Mtpa capacity) has

a clinker-to-cement ratio of 0.760, exceeding the global average of 0.73. The carbon-intensive technology mix that enabled China’s industrialization is precisely what CBAM penalizes most heavily.

Third, CBAM’s timeline aligns with a critical domestic reform window. Between 2023 and 2027, China’s Ministry of Ecology and Environment announced three landmark CN-ETS reforms: transition to absolute emissions caps by 2027, expansion to steel and cement (2024), and gradual introduction of allowance auctioning (MEE, 2021, 2023). While China’s “dual carbon” goals provide independent momentum, the temporal alignment with CBAM invites scrutiny of how external pressure interacts with domestic reform—an analytical task this paper takes up.

C. Research Question, Scope, and Contribution

This paper addresses the central question: *How does CBAM reshape China’s carbon pricing architecture and sectoral decarbonization pathways?* We decompose this into three sub-questions: (i) what channels transmit CBAM pressure to Chinese policymakers and industry; (ii) what is the magnitude of sectoral cost exposure under current and reformed CN-ETS scenarios; and (iii) what political-economy constraints bound feasible reform trajectories.

Our empirical analysis focuses on steel and cement during 2024–2027. Methodologically, we combine Global Energy Monitor plant-level data with a CBAM liability model derived directly from Regulation EU 2023/956.

This paper makes three contributions: (i) an empirically calibrated assessment of CBAM’s first-year implementation, moving beyond pre-implementation modelling; (ii) a three-channel transmission framework linking CBAM design to CN-ETS reform; and (iii) a clarification of the climate club model’s limits in non-OECD contexts.

The remainder of the paper proceeds as follows. Section II reviews the literature and identifies gaps. Section III describes data and methods. Section IV analyses CBAM design and transmission channels. Section V presents empirical results. Section VI examines industrial and policy responses. Section VII discusses implications. Section VIII concludes.

II. LITERATURE REVIEW AND THEORETICAL FRAMEWORK

A. Climate Clubs and the CBAM as a De Facto Club

Nordhaus’s (2015) climate club model proposes that a coalition adopting a common internal carbon price and imposing *uniform* penalty tariffs on non-members can sustain stable high-abatement coalitions. A growing literature examines whether the EU’s CBAM operationalises this logic. Tagliapietra and Wolff (2021) and Szulecki et al. (2022) characterise CBAM as a “de facto” climate club, yet emphasise that its incentive structure diverges from Nordhaus’s stylised model: rather than excluding non-members through uniform tariffs, CBAM’s deduction clause creates a fiscal pull toward domestic carbon pricing. More fundamental critiques question whether the club mechanism captures the actual politics of climate cooperation. Falkner (2016) and Falkner,

Nasiritousi, and Reischl (2022) argue that minilateral clubs face severe legitimacy and feasibility constraints, and that “the most stringent climate club proposals are focused on overcoming the free-riding problem rather than addressing distributive conflict” (Falkner et al., 2022). Aklin and Mildemberger (2020) push this further, arguing that climate inaction reflects domestic distributive conflict, not international free-riding—governments adopt climate policies regardless of others’ actions, suggesting external sanctions target the wrong binding constraint. Böhringer et al. (2022) and Mehling et al. (2019) converge on a similar diagnosis from a design perspective: real-world border adjustments must accommodate vastly heterogeneous national circumstances, generating administrative and equivalence challenges that the homogenous-region Nordhaus framework abstracts away. This paper tests whether these heterogeneities undermine the climate club’s self-enforcing logic in the Chinese context.

B. Two-Level Games in Authoritarian Climate Governance

Putnam’s (1988) two-level game framework, in which negotiators simultaneously bargain at international and domestic tables constrained by overlapping “win-sets,” has become a standard lens for analysing climate diplomacy (Bailer, 2012). Its application to China requires substantial adaptation. The original framework presumes electoral ratification, whereas China’s domestic “Level II” game is constructed through bureaucratic competition among ministries (MEE, NDRC, MIIT, MOFCOM), provincial fiscal interests, and state-owned enterprise lobbying (Feng, 2021; Gilley, 2012). Recent work by Chen and Lo (2025) on “authoritarian environmentalism 2.0” demonstrates that market-based instruments such as ETS reinforce hierarchical state control rather than constraining it—suggesting that international pressure interacts with Chinese policy through intra-state distributive bargaining rather than legislative veto. This reframing is consequential: external instruments such as CBAM enter Chinese policy debates not by changing aggregate national costs but by reshaping the relative bargaining position of pro-reform actors within the bureaucracy. We operationalise this logic by examining how CBAM’s differential cost impacts across technologies and provinces alter the domestic win-set for CN-ETS reform.

C. Carbon Pricing Convergence and ETS Diffusion

The literature on ETS diffusion offers two contending expectations. The optimistic view—underlying linkage proposals from the EU–Switzerland model—anticipates convergence of design as jurisdictions emulate the EU template. The empirical record is more sobering: Wettestad and Gulbrandsen (2018) document that ETS diffusion has produced design *divergence* more often than convergence, as local political-economy factors override technocratic emulation. China’s case illustrates this divergence. Stoerk, Dudek, and Yang (2019) warned at CN-ETS launch that intensity-based caps with ex-post adjustment could undermine price signals—a critique now being addressed through the announced 2027 transition to absolute caps. Cui et al. (2021)

provide firm-level evidence that even China’s regional pilots reduced emissions by 16.7% without significant output loss, suggesting that low headline prices need not imply policy ineffectiveness. Dai and Pollitt (2025), drawing on extensive document and interview evidence, conclude that CN-ETS expansion is constrained primarily by MRV bottlenecks rather than political will, and that CBAM imposes an effective 2034 deadline for full CN-ETS maturity. Together, these studies suggest that CBAM does not produce convergence directly; it alters the political-economy calculus of jurisdictions whose pre-existing reform trajectories were already underway. This paper quantifies the magnitude of this “accelerator effect” by comparing CN-ETS reform timelines before and after CBAM’s announcement.

D. Empirical Evidence on CBAM’s Post-Implementation Effects

Until 2024, almost all CBAM impact assessments relied on pre-implementation modelling. Post-implementation empirical work remains sparse but is rapidly accumulating, with two methodologically distinct studies forming the baseline for our analysis. Li et al. (2026), using a global multi-regional input-output (GMRIO) framework, estimate that China’s CBAM-induced export losses rise from EUR 0.78 billion under current scope to EUR 11.14 billion under full expansion, and find that CN-ETS can offset 30–60% of these losses depending on price scenarios and CBAM coverage. Pan et al. (2025) couple GMRIO with the GCAM-China integrated assessment model at provincial resolution and report offset rates of 42% (DICE model) to 49% (ECE scenario) for iron and steel exports specifically, with over 65% of provincial costs concentrated in Hebei, Jiangsu, Shanxi, Liaoning, and Zhejiang. The two estimates fall within overlapping ranges, but the divergence in headline figures reflects substantively different assumptions: Li covers all CBAM sectors with static GMRIO and price-scenario variation, while Pan focuses on ISAP with dynamic modelling that captures EAF technology penetration as a mitigation channel. The two studies are best read as complementary rather than competing, but neither traces the political-economy mechanism by which CBAM pressure becomes domestic CN-ETS reform.

Complementary evidence reinforces the picture: Magacho, Espagne, and Godin (2024) establish the heterogeneity of CBAM exposure across EU trade partners; Wang et al. (2024) estimate that CBAM reduces Chinese steel exporters’ profits by 58% even as environmental benefits remain marginal at the sectoral level; Clausen et al. (2025) document empirical instances of CBAM-induced policy diffusion, including China’s March 2025 expansion of CN-ETS to CBAM-covered sectors. The divergence between Li et al. (2026) and Pan et al. (2025) highlights a broader methodological limitation: static CGE models capture aggregate cost exposures but miss dynamic technology adjustment, while dynamic IAMs incorporate mitigation channels but rely on sector-average emission factors. This paper addresses both limitations by using plant-level data to model technology

heterogeneity within a static liability framework calibrated to post-implementation policy parameters.

E. Research Gaps and Analytical Framework

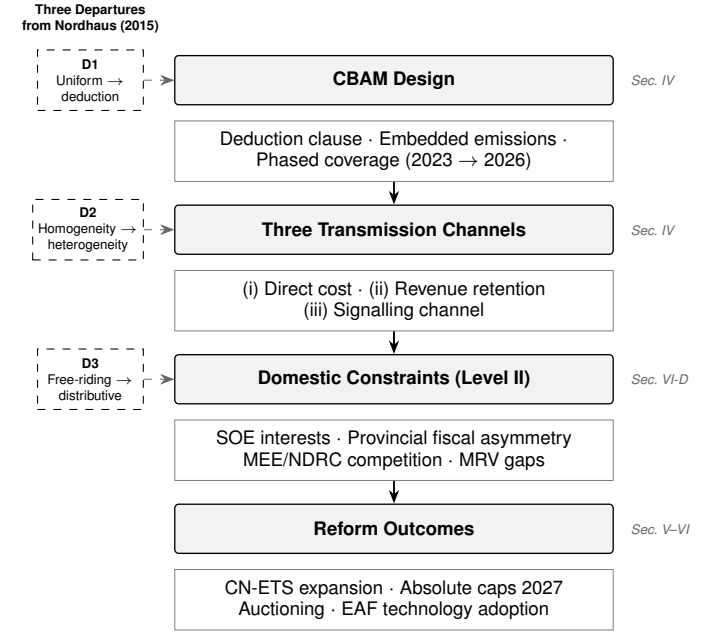


Fig. 1. Analytical framework. CBAM’s design (Stage 1) generates three transmission channels (Stage 2) that interact with China’s domestic political-economy constraints (Stage 3) to shape CN-ETS reform outcomes (Stage 4). Three departures from Nordhaus’s stylised climate club (left, D1–D3) define the paper’s theoretical contribution; Stage 3 constraints operationalise Putnam’s Level II.

Synthesising the above literature, we identify three critical departures of CBAM from Nordhaus’s stylised climate club model that have not been jointly theorised or empirically tested (summarised in Fig. 1). *First*, CBAM’s deduction clause replaces Nordhaus’s uniform tariff with a differentiated, emission-intensity-dependent liability that endogenises the choice of domestic carbon pricing—transforming the club mechanism from exclusionary tariff to gravitational fiscal pull. *Second*, CBAM’s non-member targets exhibit vast industrial-structure heterogeneity that Nordhaus’s homogenous-region C-DICE framework cannot capture; China’s BF-BOF dominance and provincial concentration of CBAM exposure (Pan et al., 2025) generate response capacities and constraints fundamentally different from those of EU member states. *Third*, the binding constraint on CN-ETS reform is not international free-riding but domestic distributive conflict among provincial governments, SOEs, and central reformers (Aklin and Mildenberger, 2020; Cui et al., 2021; Dai and Pollitt, 2025)—a constraint Nordhaus’s external-enforcement logic does not address.

These departures motivate three core research gaps that this paper addresses:

- 1) **Mechanism gap:** Existing CGE and IAM studies compute equilibrium offset outcomes without mapping the causal transmission mechanism from CBAM external pressure to specific CN-ETS reform measures.

- 2) **Heterogeneity gap:** Sector-average emissions modelling obscures the plant-level technology heterogeneity that drives the BF-BOF-EAF cost differential central to China’s adjustment pathway.
- 3) **Context gap:** No study has systematically integrated domestic distributive conflict into an empirical assessment of CBAM’s impact on non-OECD carbon pricing systems.

To address these gaps, we develop a transmission-channel framework comprising the direct cost channel, the revenue retention incentive, and the signalling channel (Section IV), and apply it to plant-level data on China’s steel and cement sectors (Sections V–VI). The framework operationalises Putnam’s two-level logic in the authoritarian-bureaucratic context and integrates the distributive-conflict reading of climate politics into a tractable empirical assessment.

III. METHODOLOGY AND DATA

A. Data Sources

Analysis draws on four primary datasets. Plant-level steel capacity and technology data come from the GEM Iron & Steel Tracker (GEM, 2026), comprising 458 operating plants with unit-level BOF/EAF technology classification, provincial location, and production capacity. Cement capacity data come from the GEM Cement and Concrete Tracker (GEM, 2025), covering 1,016 operating plants with clinker and cement capacity disaggregated by plant type (integrated vs. grinding-only). Sectoral production benchmarks draw on IEA (2023) and WorldSteel (2024). Carbon pricing data for the CN-ETS are sourced from MEE official publications (MEE, 2021, 2023); EU ETS price ranges reflect 2024–2025 market data.

B. Emission Factor Derivation

Three sector-specific emission factors are applied throughout:

BF-BOF steel: 2.10 tCO₂/t crude steel, representing Scope 1 direct process emissions. WorldSteel (2025) reports a global BF-BOF Scope 1+2 intensity of 2.34 tCO₂/t (WorldSteel, 2025), with Scope 2 forming a minor share given coal’s dominance over electricity in the process energy mix.

EAF steel: 0.10 tCO₂/t crude steel (Scope 1, **CBAM-applicable only**). Under Regulation EU 2023/956, Scope 2 (indirect) electricity emissions are **excluded** from CBAM certificate liability for steel; only Scope 1 direct process emissions apply. EAF Scope 1 of 0.10 tCO₂/t reflects direct process emissions (WorldSteel, 2025). For reference, WorldSteel (2025) reports global scrap-EAF total intensity of 0.69 tCO₂/t including Scope 2 (WorldSteel, 2025). This total figure is reported for technology comparison only and does *not* affect CBAM liability under Regulation EU 2023/956. Note that applying China’s national grid factor (0.581 kgCO₂/kWh; Fig. 2) to a typical EAF electricity consumption of approximately 580 kWh/t steel yields a Scope 2 term of only ≈ 0.34 tCO₂/t, implying a China-specific total of ≈ 0.44 tCO₂/t—substantially below the WorldSteel global figure of 0.69. The difference reflects WorldSteel’s use of

a global-average grid mix and higher assumed electricity intensity; the 0.69 figure is retained here as the published global benchmark and is not used in any liability calculation. China’s power generation mix and associated grid emission factor are shown in Fig. 2.

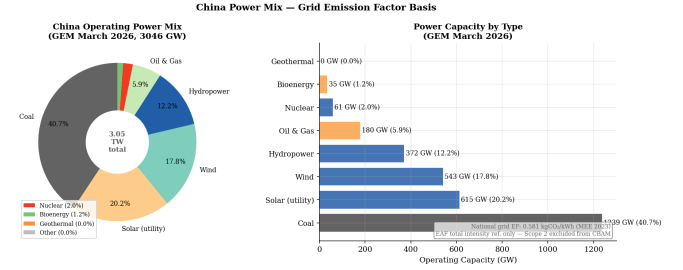


Fig. 2. China’s power generation mix and associated grid emission factor (0.581 kgCO₂/kWh) (MEE, 2023). Shown for reference; WorldSteel (2025) global scrap-EAF total intensity is 0.69 tCO₂/t including Scope 2 (WorldSteel, 2025). Scope 2 is excluded from CBAM liability for steel under Regulation EU 2023/956 (Scope 1 only: 0.10 tCO₂/t).

Cement: 0.627 tCO₂/t finished cement, derived from the GEM-observed clinker-to-cement ratio of 0.760 multiplied by the IPCC (2006) clinker emission factor of 0.825 tCO₂/t clinker (calcination: 0.525; coal combustion kiln heat: 0.300) (IPCC, 2006). This figure covers **Scope 1 calcination and combustion emissions only**. Commission Implementing Regulation (EU) 2023/1773, Annex III requires that Scope 2 indirect electricity emissions be included for cement under CBAM; at China’s national grid factor of 0.581 kgCO₂/kWh and typical electricity consumption of 90–110 kWh/t cement (IEA, 2023), the omitted Scope 2 term is approximately 0.052–0.064 tCO₂/t (an understatement of roughly 8–10%). This study retains the Scope 1-only figure for consistency with the plant-level GEM dataset and for comparability with analogous estimates in the literature; the omission is acknowledged as a conservative bias in the cement liability estimates reported in Table II.

C. CBAM Liability Model

Net CBAM liability per tonne of exported product i is computed as:

$$\text{CBAM Liability}_i = \max(0, EF_i \times (P_{\text{EU}} - P_{\text{CN}})) \quad (1)$$

where EF_i is embedded emissions intensity (tCO₂/t), P_{EU} is the prevailing EU ETS allowance price (EUR/tCO₂), and P_{CN} is the verified CN-ETS carbon cost paid in the country of origin. Steel and cement sectors entered CN-ETS coverage in 2024; P_{CN} therefore reflects the prevailing CN-ETS allowance price for covered facilities, currently EUR 10–13/tCO₂. Sensitivity analysis spans two EU ETS price scenarios (EUR 65 and EUR 80/tCO₂) and five CN-ETS price states.

D. Scope and Limitations

Aggregate CBAM exposure estimates are bounded to current CBAM sectoral scope (HS Chapter 72 iron and steel; cement clinker); downstream embedded emissions in

manufactured goods are noted but not modelled. Provincial export volumes are approximated from capacity shares rather than trade flow data, introducing uncertainty in absolute cost estimates.

P_{CN} uniformity. Article 9(1)–(3) of Regulation (EU) 2023/956 requires that deductible carbon costs be effectively paid at the specific installation, not the national ETS average price. This study assigns $P_{CN} = \text{EUR } 13/\text{tCO}_2$ uniformly to all GEM-tracked steel and cement plants post-2024, without filtering for plant-level CN-ETS coverage status, compliance thresholds, or effective payment. Plants outside CN-ETS compliance obligations—including small facilities below sectoral thresholds or in non-covered sub-categories—would have a deductible $P_{CN} = 0$, implying that the actual aggregate offset may be lower than Table II suggests. This is treated as a limitation; plant-level coverage filtering is deferred to future work with administrative MRV data.

Clinker emission factor. The calcination component of the clinker EF (0.525 tCO₂/t clinker) implies a CaO fraction of approximately 67.6%, above the IPCC (2006) Tier 1 default of 0.507 tCO₂/t clinker (CaO fraction $\approx 65\%$). A China-specific clinker composition study justifying this deviation is not available in the open literature; the figure is sourced from IEA (2023) sectoral benchmarks. The difference from the IPCC default is approximately 3.5% in the calcination term, or 1.8% in the total clinker EF, well within the sensitivity range reported in Section V-B.

Causality is a further limitation: the correlation between CBAM implementation and CN-ETS reform acceleration is consistent with the analytical framework in Section II.E, but formal causal identification requires counterfactual evidence beyond the scope of this study. Findings are interpreted as associations informed by theoretical priors, not demonstrated causal effects. These parameters feed directly into the liability model applied in Section V.

IV. CBAM DESIGN AND TRANSMISSION MECHANISMS

A. Policy Design and Transmission Logic

CBAM (Regulation EU 2023/956) entered its definitive phase on 1 January 2026, following a transitional reporting-only phase from 1 October 2023 through 31 December 2025 (EU, 2023). EU importers of products listed in Annex I—iron and steel, aluminium, cement, fertilisers, electricity, and hydrogen—must now register as authorised CBAM declarants, surrender CBAM certificates by 31 May each year against the preceding calendar year’s verified embedded emissions, and reconcile any outstanding liability with EU customs. Certificate prices track the weekly auction-clearing price of EU ETS allowances, ensuring price equivalence between domestic compliance and border-adjusted imports.

Embedded emissions are defined as Scope 1 direct emissions for steel and hydrogen, with Scope 2 indirect electricity-related emissions added for cement and fertilisers under the Regulation’s definitional framework. This study’s cement liability calculations use Scope 1 only; see Section III-B for the quantified impact of this divergence. Where verified installation-level data are unavailable, default values

published by the European Commission apply; these are typically benchmarked at the worst-decile producers globally, generating a strong incentive for facility-level monitoring, reporting, and verification (MRV).

CBAM’s strategic significance for China rests on Article 9 of the Regulation, which permits the deduction of carbon costs already paid in the country of origin from CBAM liability on a euro-for-euro basis, provided those costs are verifiable and meet equivalence criteria specified in implementing regulations. This deduction clause is the fulcrum of the mechanism’s interaction with Chinese carbon policy: each yuan of effective CN-ETS compliance cost converts what would otherwise be a transfer to the EU budget into revenue retained domestically. The parallel free-allocation phase-out under EU ETS Annex I—running 2026–2034 in linear annual decrements—calibrates CBAM coverage to internal carbon-cost equalisation rather than additional protection, a design feature critical to WTO defensibility.

Three transmission channels follow from this architecture and are tested empirically in Sections V and VI:

- The **direct cost channel**: CBAM raises the landed cost of Chinese exports in proportion to embedded emissions intensity, with the per-tonne liability given by Eq. 1. The competitiveness implication scales with the Chinese carbon-intensity disadvantage relative to EU producers operating under stringent ETS prices.
- The **revenue retention incentive**: the Article 9 deduction transforms domestic carbon pricing from a pure compliance cost into a fiscal-substitution instrument. Each marginal increase in P_{CN} shifts an equivalent quantum of revenue from EU coffers to Chinese ones, bounded above by P_{EU} . This converts a passive trade-policy exposure into an active strategic incentive for CN-ETS reform.
- The **signalling channel**: the publicly observable P_{EU} functions as an external benchmark for CN-ETS ambition. Even where direct deduction logic is incomplete (e.g., during sectoral phase-in delays), the price gap is read by domestic policy actors—MEE, NDRC, sectoral associations—as an indicator of how far the reform agenda must travel.

A fourth, procedural channel emerges indirectly from MRV requirements: CBAM’s documentation standards effectively impose ISO 14064-aligned carbon accounting on Chinese exporters as a precondition for accessing facility-specific (rather than punitive default) emission factors. From the firm’s perspective this is sunk compliance cost; from the system’s perspective it is a public-good investment in carbon-market infrastructure that complements CN-ETS expansion—an interaction Dai and Pollitt (2025) document as a binding bottleneck on CN-ETS maturity.

B. Comparative Architecture: EU ETS and CN-ETS

CBAM’s revenue-retention logic operates entirely through the price gap between the two emissions trading systems on either side of the Article 9 deduction. Understanding the

structural drivers of that gap requires direct comparison of EU ETS and CN-ETS architectures, summarised in Table I.

The EUR 40–65 / tCO₂ price gap is structurally generated by three differences. *First*, EU ETS imposes an absolute cap whose annual contraction is mechanically priced into the forward curve, whereas CN-ETS intensity benchmarking allows allowance volumes to expand with output, dampening upward price pressure (Dai and Pollitt, 2025; Stoerk et al., 2019). *Second*, EU allocation has been steadily migrating from free to auctioned, raising effective compliance costs in line with the cap, while Chinese steel and cement entered CN-ETS in 2024 with generous free allocation to ease the transition. *Third*, EU ETS coverage spans the bulk of stationary-combustion and industrial emissions, while CN-ETS covered only the power sector through 2023 and remains in mid-expansion.

These structural features are the binding constraint on convergence. Within the two-level games framework adopted in Section II-B, they define the upper limit of how rapidly CN-ETS can be tightened without compromising sectoral employment and energy-security imperatives (Nahm and Urpelainen, 2021). CBAM enters this domestic equilibrium as an external incentive that shifts the cost–benefit calculation of reform—not by mandating convergence but by attaching a measurable fiscal-leakage cost to non-convergence.

C. Sectoral Coverage Logic and Expansion Trajectory

The sectors selected for CBAM Annex I coverage—cement, iron and steel, aluminium, fertilisers, electricity, and hydrogen—share three features that define the mechanism’s analytical logic. *First*, each is highly carbon-intensive, with embedded emissions in the range of 0.6–2.1 tCO₂ per tonne of finished product. *Second*, each has documented exposure to carbon leakage under the EU ETS prior to CBAM, evidenced by free-allocation eligibility and trade-intensity metrics maintained by the European Commission. *Third*, each is technologically mature enough to permit reasonably standardised emission-factor methodologies, securing administrative feasibility within a customs-based system.

Hydrogen and electricity are forward-looking inclusions. Hydrogen exposure is currently small but anticipated to grow as the EU’s Renewable Energy Directive recast (RED III) drives demand for low-carbon hydrogen imports; electricity inclusion addresses cross-border grid leakage from non-EU jurisdictions adjacent to EU member states (Western Balkans, Türkiye), and is of limited direct relevance for China.

The phase-in trajectory is designed for parallelism with EU ETS free-allocation reduction. Free allowances for CBAM-covered EU sectors decline from 100% in 2025 to 0% in 2034 in linear annual decrements; CBAM certificate obligations rise correspondingly, ensuring that European producers always face the full carbon cost of their emissions and that Chinese exporters face an equivalent cost on the residual gap not covered by CN-ETS payments. This calibration is critical both to WTO defensibility and to the mechanism’s framing as equalisation rather than protection.

Two expansion vectors are particularly consequential for China. The first is the 2025–26 mandated review of CBAM scope, widely expected to recommend extension to organic chemicals, polymers, and selected downstream manufactured goods (refined steel products, cement-based construction materials). Li et al. (2026) project that this expansion would lift China’s aggregate CBAM exposure from EUR 0.78 billion under current pilot scope to EUR 11.14 billion under full coverage—a 14× increase reflecting both higher product volumes and the deeper integration of CBAM-regulated inputs into Chinese manufacturing exports. The second vector is the expansion of indirect (Scope 2) emissions coverage. Currently included only for cement and fertilisers, the inclusion of Scope 2 for steel would substantially raise the per-tonne liability of Chinese producers reliant on coal-dominant grids; the elasticity of this exposure to provincial grid factors is quantified in Section V-D.

D. Strategic Foresight: CBAM as a Precedent for Global Border Carbon Adjustments

CBAM is no longer a single-jurisdiction risk for Chinese exporters. The United Kingdom has confirmed that its own CBAM will enter force on 1 January 2027, covering cement, fertilisers, hydrogen, and iron and steel under primary legislation enacted in the Finance Act 2026, with secondary legislation under technical consultation through mid-2026 (HMRC, 2026). The UK design parallels the EU regime but excludes electricity and operates with a single global default value per CN code in its first year, simplifying administration at the cost of country-of-origin differentiation. The UK and EU concluded an agreement in principle in May 2025 to link their respective ETS systems and explore mutual CBAM exemption, which would consolidate a transatlantic European carbon-pricing zone with implications for Chinese MRV-equivalence negotiations.

In the United States, the bipartisan Foreign Pollution Fee Act of 2025 (S. 1325) was reintroduced in April 2025 by Senators Cassidy and Graham, proposing tiered tariffs on imports of iron, steel, cement, glass, fertilisers, hydrogen, and selected critical-mineral and battery inputs (Cassidy and Graham, 2025). Unlike the EU and UK mechanisms, the US proposal does not impose a domestic carbon price and does not formally credit foreign carbon costs; it computes pollution-intensity differentials relative to US production and applies ad valorem tariffs accordingly. This design departs from the price-equivalence logic of EU CBAM and would, if enacted, expose Chinese producers to a parallel charge with no offset pathway through CN-ETS payments. Whether the bill advances is uncertain—prior versions stalled in 2023–24—but its bipartisan support and explicit geopolitical framing against China give it material policy weight (Rasool et al., 2024). The accompanying PROVE IT Act, which would direct the U.S. Department of Energy to study comparative carbon intensities, advanced from committee in the 118th Congress and provides the data infrastructure on which any subsequent border-adjustment regime would rely.

TABLE I
STRUCTURAL COMPARISON OF EU ETS AND CN-ETS (2024–2025)

Dimension	EU ETS	CN-ETS
Year established	2005 (Phase I); Phase IV running 2021–2030	2021 (national); regional pilots from 2013
Sectoral coverage	Power, energy-intensive industry, intra-EEA aviation, maritime (from 2024)	Power generation; steel and cement (from 2024)
Cap design	Absolute, declining; Linear Reduction Factor 4.3% (2024–27), 4.4% (2028–30)	Intensity-based; transition to absolute cap announced for 2027
Allocation method	Hybrid: auctioning (~57%); free benchmark-based allocation declining via cross-sectoral correction factor	Free allocation against intensity benchmarks; auctioning pilots in select provinces
Price (2024–25)	EUR 50–80 / tCO ₂	EUR 10–13 / tCO ₂ (CNY 80–105)
MRV regime	Accredited third-party verifiers under ISO 14065; central registry	Sectoral verifiers under MEE supervision; provincial registries integrating into national
Compliance penalty	EUR 100/tCO ₂ + missed-allowance surrender + name disclosure	Administrative penalty typically 5–10× market price; reform pending
Linkage capability	Linked with Swiss ETS since 2020	No active linkage; bilateral cooperation under exploration
Revenue use	Member states (climate purposes ≥50%); Innovation Fund; Modernisation Fund	MEE; sectoral allocations; reform direction toward general fiscal absorption
Offsets	Limited international credits, no longer accepted post-Phase III	CCER (China Certified Emission Reductions) reactivated 2024, capped

Canada’s federal climate plan under the Carney government has indicated active consideration of a CBAM as part of its 2030 industrial-policy package, and Australia conducted public consultations on a Carbon Leakage Review concluding in 2024. While neither jurisdiction has confirmed implementation, the cumulative direction of policy diffusion is unambiguous: the EU’s first-mover act has shifted the equilibrium of acceptable trade–climate policy, and China should plan for cumulative exposure to multiple, partially overlapping border-adjustment regimes by 2028–30 rather than for the EU mechanism in isolation.

This diffusion has two strategic implications. First, the marginal value of strengthening CN-ETS rises with each additional jurisdiction adopting deduction-based CBAMs, since the same domestic carbon cost is increasingly off-settable across multiple borders. Second, the heterogeneity of mechanism designs—deduction-based (EU, UK) versus differential-based (US)—means that no single CN-ETS reform pathway optimises against all of them simultaneously; harmonisation pressures in MRV standards and verifier accreditation will likely intensify, whether through bilateral mutual-recognition agreements or multilateral fora such as the OECD Inclusive Forum on Carbon Mitigation Approaches (IFCMA).

V. EMPIRICAL RESULTS

A. Sectoral Exposure: China’s CBAM Vulnerability

China constitutes the single largest non-EU source country for CBAM-liable exports, with exposure concentrated in steel and cement.

Steel. China’s GEM-tracked steel industry comprises 458 plants with 1,005 Mtpa of operating crude steel pro-

duction capacity—approximately 54% of global crude steel production (WorldSteel, 2024). Of this capacity, 835.7 Mtpa (83.1%) is attributable to BF-BOF operations and 169.5 Mtpa (16.9%) to EAF. China’s EAF share relative to global peers is shown in Fig. 3. China’s weighted-average CBAM-applicable emissions intensity (Scope 1) across its current production mix is $0.831 \times 2.10 + 0.169 \times 0.10 = 1.76$ tCO₂/t, vs. 0.10 tCO₂/t for a pure EAF producer—a 17.6:1 fleet-average-to-EAF intensity ratio that translates directly into the core CBAM competitiveness liability. Note that the per-tonne liability ratio between BF-BOF and EAF at EUR 80/tCO₂ is EUR 140.7 vs. EUR 6.7, or approximately 21:1, reflecting the product of the sectoral emission-factor gap and the carbon-price level.

Li et al. (2026) estimate aggregate CBAM-induced export losses rising from EUR 0.78 billion under the current pilot sectors to EUR 11.14 billion with full CBAM expansion across all covered products. Pan et al. (2025) project CBAM costs of EUR 1,157 million by 2034 for China’s iron, steel, and aluminium exports to the EU, with over 65% of the burden falling on Hebei, Jiangsu, Shanxi, Liaoning, and Zhejiang. Provincial BF-BOF capacity distribution underlying this geographic concentration is shown in Fig. 4. Actual exposure depends on export orientation; inland capacity serving domestic markets may face lower realized CBAM costs than coastal export-oriented capacity of equivalent scale.

Cement. *Note on scope:* direct CBAM exposure from cement exports to the EU is structurally limited under current-phase implementation, as cement’s low value-to-weight ratio precludes economically meaningful long-distance trade. The quantitative analysis below is therefore best understood as a forward-looking scenario relevant to future CBAM scope

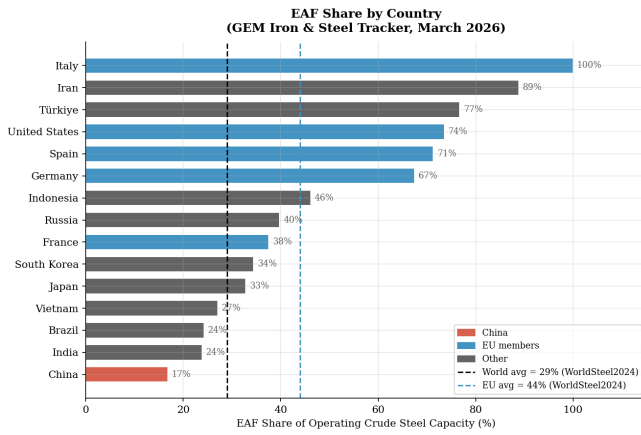


Fig. 3. EAF share of crude steel production capacity (%) by country. China's 16.9% compares with $\approx 29\%$ globally and $\approx 44\%$ in the EU, and $>70\%$ in the United States (WorldSteel, 2024). Data: GEM Iron & Steel Tracker, March 2026 (GEM, 2026).

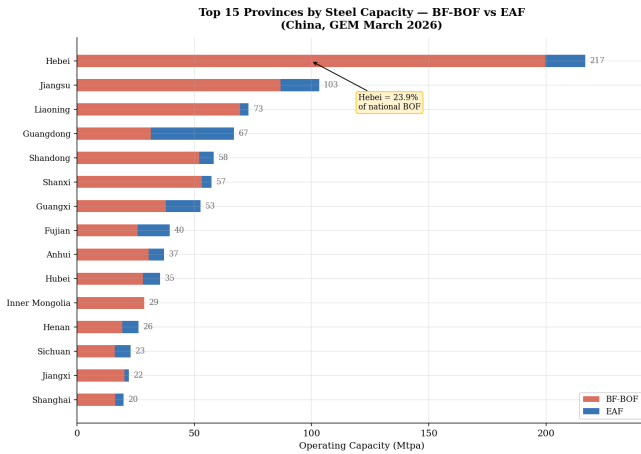


Fig. 4. Operating BOF and EAF capacity (Mtpa) by province, China. Hebei (199.7 Mtpa BOF), Jiangsu (86.8), Liaoning (69.5), Shanxi (53.2), and Shandong (52.0) collectively account for 55.2% of national BOF capacity. Data: GEM Iron & Steel Tracker, March 2026 (GEM, 2026).

expansion—particularly any extension to precast construction materials or cement clinker—rather than as a measure of near-term fiscal exposure.

China accounts for approximately 50–51% of global cement output (GCCA, 2024). GEM identifies 1,016 operating plants with 1,772 Mtpa cement capacity and 1,346 Mtpa clinker capacity. At the IPCC (2006) clinker emission factor of 0.825 tCO₂/t clinker (IPCC, 2006), Chinese integrated plants emit approximately 0.627 tCO₂/t of finished cement (Scope 1; see Section III-B). The absence of large direct flows does not eliminate exposure: cement is embedded in higher-value construction products (precast elements, ready-mix concrete) whose Scope 3 emissions could fall within future CBAM reviews, and China's cement-export profile is heavily oriented toward emerging-market construction markets where any climate-club expansion would have second-order trade implications.

Aluminium. Aluminium is listed in CBAM Annex I but

is excluded from the quantitative analysis in this paper; the analysis that follows is scoped to steel and cement only.

B. CBAM Cost Quantification

Applying Eq. 1 with emission factors from Section III-B, and with steel and cement carrying $P_{CN} \approx \text{EUR } 10\text{--}13/\text{tCO}_2$ following their 2024 CN-ETS inclusion, net CBAM liability per tonne is given in Table II and visualized in Fig. 5.

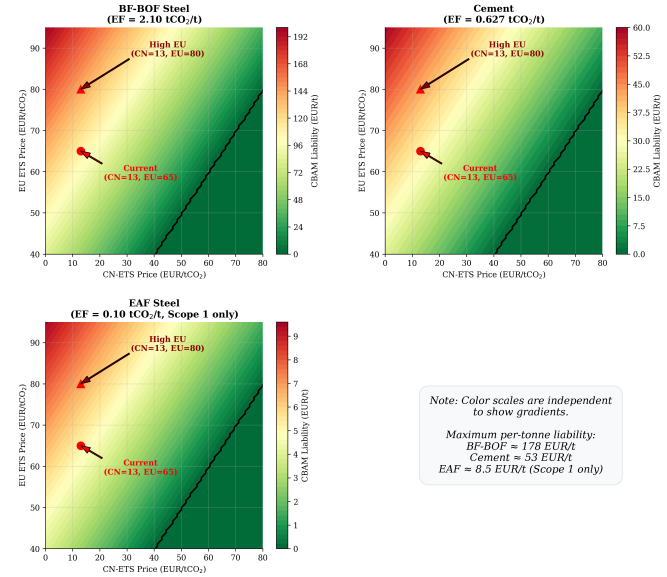


Fig. 5. Heatmap of net CBAM liability (EUR/t) across EU ETS price scenarios and CN-ETS coverage levels for BF-BOF steel, EAF steel, and cement. Derived from Table II.

Aggregate fiscal exposure. Aggregate CBAM cost depends jointly on per-tonne liability, EU-bound export volumes by sector, and the EU ETS price scenario. Under the current pilot scope and current carbon-price gap, Li et al. (2026) and Pan et al. (2025) jointly bound the annual fiscal exposure at EUR 0.78–1.16 billion, with the latter rising to EUR 1.16 billion by 2034 in their projection. Under the post-2026-review scope expansion, Li et al. (2026) project a ceiling of EUR 11.14 billion. We treat these figures as outer bounds rather than central estimates, given heterogeneity in underlying methodology: Pan et al. (2025) construct estimates from province-level capacity data within a dynamic IAM, while Li et al. (2026) employ static GMRIO that captures sectoral input–output linkages including indirect downstream exposure absent from the former.

Provincial burden distribution. The provincial allocation of this fiscal exposure is highly skewed. Pan et al. (2025) attribute over 65% of projected CBAM costs to five provinces—Hebei, Jiangsu, Shanxi, Liaoning, and Zhejiang—broadly tracking BF-BOF capacity distribution (Fig. 4). Hebei alone bears an estimated 23% of the national burden under their methodology. This concentration creates an asymmetric political economy of reform: a small set of provincial governments would bear most of the visible cost, but national-level fiscal benefits from CN-ETS revenue retention accrue centrally, generating intergovernmental

TABLE II
NET CBAM LIABILITY PER TONNE OF PRODUCT BY CN-ETS COVERAGE AND PRICE SCENARIO (EUR/T)

CN-ETS Scenario	EU ETS = EUR 65/tCO ₂			EU ETS = EUR 80/tCO ₂		
	BF-BOF	EAF	Cement	BF-BOF	EAF	Cement
Not covered (pre-2024 baseline)	136.5	6.5	40.8	168.0	8.0	50.2
Covered, EUR 13/tCO ₂ (<i>current</i>)	109.2	5.2	32.6	140.7	6.7	42.1
Covered, EUR 40/tCO ₂	52.5	2.5	15.7	84.0	4.0	25.1
Covered, EUR 60/tCO ₂	10.5	0.5	3.1	42.0	2.0	12.5
Full parity (EU ETS level)	0.0	0.0	0.0	0.0	0.0	0.0

Note: Emission factors: BF-BOF 2.10 tCO₂/t Scope 1 (WorldSteel 2025 global Scope 1+2: 2.34 tCO₂/t; WorldSteel 2025); EAF **0.10 tCO₂/t Scope 1 only** (CBAM-applicable; Scope 2 excluded per Regulation EU 2023/956; global scrap-EAF total 0.69 tCO₂/t including Scope 2; WorldSteel 2025); Cement 0.627 tCO₂/t (GEM clinker ratio 0.760 × IPCC 0.825 tCO₂/t clinker; IPCC 2006). “Pre-2024 baseline” row assumes $P_{CN} = 0$; “current” row reflects CN-ETS entry of steel and cement in 2024 at EUR 13/tCO₂. Figures rounded to one decimal place.

coordination challenges that the analytical framework in Section II-B treats as an extension of Putnam’s two-level logic to subnational federal-style bargaining.

Sensitivity to emission factors. The base case uses point estimates for emission factors (BF-BOF 2.10, EAF Scope 1 only 0.10, cement 0.627 tCO₂/t). Plausible ranges around these, drawn from IEA (2023), GEM unit-level data, and IPCC (2006), are: BF-BOF 1.95–2.30, EAF Scope 1 0.08–0.12, cement 0.58–0.68 tCO₂/t. Because Eq. 1 is linear in EF_i , the per-tonne liability scales proportionally with the emission factor: a $\pm 10\%$ deviation in BF-BOF intensity translates into a $\pm 10\%$ deviation in BF-BOF liability, and similarly for cement and EAF. The qualitative ranking BF-BOF $\gg$ cement > EAF (Scope 1) is preserved across the entire plausible range, since the gaps between sectoral intensities exceed the within-sector uncertainty by an order of magnitude. The provincial grid-factor question, which matters for any future Scope 2 extension, is treated separately in Section V-D.

C. Carbon Price Gap and Convergence Scenarios

As of 2024–2025, the EU ETS price has traded at EUR 50–75/tCO₂ vs. EUR 10–13/tCO₂ for CN-ETS. Three convergence scenarios toward full CBAM offset are illustrated in Fig. 6. Li et al. (2026) find that China’s ETS helps reduce 30–60% of CBAM-induced export losses, with the offset magnitude positively correlated with CN-ETS price levels—underscoring the strategic value of domestic carbon pricing reform as a CBAM mitigation instrument.

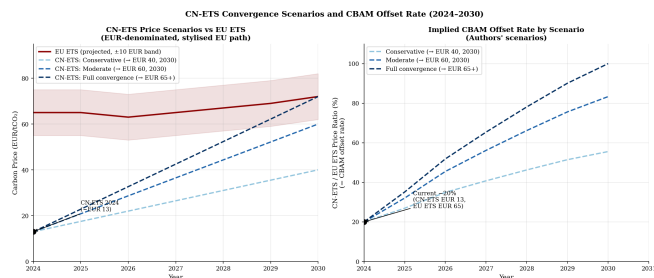


Fig. 6. CN-ETS price convergence scenarios toward EU ETS level, 2024–2035. Conservative: EUR 40/tCO₂ by 2030; Moderate: EUR 60/tCO₂ by 2030; Full convergence: price parity with EU ETS. Authors’ scenarios.

Counterfactual: reform without CBAM. Attribution of CN-ETS reforms to CBAM pressure requires a counterfactual. Two observations bound it. First, CN-ETS reform announcements predate CBAM’s adoption: the national ETS was announced in 2017, launched in 2021, and the dual-carbon goals (peak by 2030, neutrality by 2060) were formalised in September 2020—all before CBAM’s October 2023 transitional-phase entry. Second, however, the timing of the announced 2024 sectoral expansion (steel, cement) and the 2027 absolute-cap target is closely calibrated to CBAM’s definitive-phase entry in January 2026. The most defensible reading—consistent with the framework in Section II-B and with Dai and Pollitt (2025)—is that CBAM functioned as an accelerator and sequencing constraint on a reform trajectory that would have proceeded more gradually in its absence, rather than as the originating cause of reform. This interpretation acknowledges policy diffusion as multi-determined and avoids the over-attribution that single-instrument analyses can encourage.

Policy diffusion timeline. The reform trajectory traces as follows:

- **2013:** Pilot ETS programmes launched in Beijing, Shanghai, Guangdong, Tianjin, Shenzhen, Hubei, and Chongqing.
- **2017:** NDRC announces national ETS in the power sector.
- **September 2020:** Xi Jinping announces dual-carbon goals at the UN General Assembly.
- **February 2021:** National CN-ETS launched, covering power generation only (MEE, 2021).
- **October 2023:** EU CBAM transitional phase begins.
- **2024:** Steel and cement enter CN-ETS; CCER market reactivated.
- **January 2026:** EU CBAM definitive phase enters force.
- **2025–26 (planned):** Aluminium enters CN-ETS.
- **2027 (planned):** CN-ETS transition from intensity-based to absolute cap; MEE auctioning pilot.
- **January 2027:** UK CBAM enters force (HMRC, 2026).
- **2030–35:** Convergence trajectory toward EU ETS price level conditional on cap stringency and offset policy.

The clustering of activity in 2024–27—bracketing CBAM’s definitive-phase entry—is the empirical signature

most consistent with the signalling-channel mechanism (Section IV-A). It is not, however, evidence of single-cause CBAM-driven reform: the dual-carbon timeline would have produced sectoral expansion in roughly the same window even absent EU pressure, though probably with more relaxed initial caps and slower auctioning.

D. Robustness Checks

The point estimates above rest on parameter assumptions that warrant sensitivity testing in three directions: alternative grid factors, EU ETS price volatility, and limits on causal identification.

Alternative grid factors and the Scope 2 question. The MEE national grid emission factor (0.581 kgCO₂/kWh) used in Section III-B masks substantial regional heterogeneity. Provincial grid factors range from approximately 0.30 kgCO₂/kWh in hydropower-dominant Yunnan and Sichuan to 0.85+ kgCO₂/kWh in coal-dominant North China and Northwest grid regions. Under the current Regulation EU 2023/956 scope, this regional variation does *not* affect steel CBAM liability, since Scope 2 is excluded. It does, however, affect two adjacent quantities: (i) the technology-comparison total intensity for EAF (0.27 in hydropower regions vs. 0.59 tCO₂/t in coal-dominant regions), which informs the policy signalling channel even where it does not directly enter liability; and (ii) the latent exposure of Chinese steel under any future CBAM revision that brings Scope 2 into scope.

EU ETS price volatility. The base scenarios use point estimates of EUR 65 and EUR 80/tCO₂ for P_{EU} . Historical 2024–25 EU ETS prices ranged from EUR 50/tCO₂ (mid-2024 trough) to EUR 85/tCO₂ (early-2023 peak). At P_{EU} = EUR 50, BF-BOF liability under current P_{CN} falls to EUR 77.7/t (vs. EUR 140.7 at EUR 80); at P_{EU} = EUR 100, it rises to EUR 182.7/t. The roughly 2× range across plausible price scenarios reinforces that CBAM exposure is materially uncertain even before considering CN-ETS reform dynamics, and underscores the value of the deduction clause in dampening this volatility from the perspective of Chinese exporters: any increase in P_{EU} that lifts CBAM liability simultaneously raises the marginal value of P_{CN} to MEE, partially internalising the volatility.

Causal identification caveats and selection effects. As noted in Section III, the correlation between CBAM implementation and CN-ETS reform acceleration is consistent with the framework in Section II-E but does not constitute formal causal identification. Counterfactual evidence—e.g., a comparable jurisdiction subject to CBAM exposure but without independent decarbonisation goals—does not exist at sufficient scale to support difference-in-differences estimation. Findings are therefore interpreted as theoretically informed associations, not demonstrated causal effects. A second selection issue arises in the CBAM-exposure analysis itself: the model considers EU-bound exports rather than total Chinese steel and cement output, implicitly assuming that EU-bound product mix matches national-average emissions intensity. If EU-bound exporters disproportionately

use cleaner production technologies (a plausible response to CBAM signalling), our point estimates would over-state actual exposure; we therefore treat the figures here as upper bounds in this respect. Finally, Pan et al.’s (2025) provincial allocation uses capacity shares as a proxy for trade-flow shares, which may misallocate exposure if certain provinces specialise disproportionately in domestic-market or non-EU export production. We flag these as material limitations and recommend that future work integrate customs-level export microdata to disentangle the production-mix from the destination-mix effects.

VI. INDUSTRIAL AND POLICY RESPONSE

A. Steel Adjustment Pathways

China’s steel sector presents the most acute CBAM exposure, combining the world’s highest absolute production volume with an emissions-intensive technology mix geographically concentrated in provinces where decarbonization pathways are structurally constrained (Fig. 4).

The EUR 130.0/t per-tonne CBAM liability differential between BF-BOF and EAF at EUR 65/tCO₂ (pre-2024 baseline; Table II) can be partially neutralized through: (i) product switching, (ii) EAF capacity expansion in coastal provinces, and (iii) process adjustment within BF-BOF via scrap charging or hydrogen injection.

The scrap supply constraint is a binding limit on transition pace. China’s steel fleet is relatively young—reflecting the rapid capacity build-out of the 2000s—and domestic scrap availability remains limited. Chen et al. (2023) project EAF deployment will concentrate in Jiangsu, Guangdong, and Sichuan, while CCUS deployment will concentrate in Hebei, Shandong, Liaoning, and Jiangsu. Scrap availability is projected to rise substantially through 2030–2035 as end-of-life scrap from early 2000s infrastructure reaches recovery (IEA, 2023), enabling a medium-term decarbonization wave that CBAM may accelerate—consistent with, though not demonstrated by, the fiscal incentive channel in Section IV-A.

B. Cement Decarbonization Constraints

Cement decarbonization is constrained by the thermodynamic irreducibility of calcination: $\approx 63\%$ of direct emissions arise from $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$, which cannot be eliminated through fuel switching or energy efficiency alone. GEM identifies 1,016 operating plants with 1,772 Mtpa cement capacity; 782 are fully integrated and 214 are grinding-only. The clinker-to-cement ratio of 0.760 exceeds the global average of 0.73 (GCCA, 2024). Translating the cost exposures modelled in Section V-B into investment imperatives, CCS deployment status and CBAM-implied breakeven economics are shown in Fig. 7.

The CBAM cost floor (Section V-B) provides a partial economic signal for CCS investment. With a post-combustion capture rate $\eta = 0.90$ (Homaei et al., 2026), the CBAM saving per tonne of cement from deploying CCS is:

$$\Delta_{\text{CBAM}} = EF_{\text{cement}} \times \eta \times P_{EU} = 0.627 \times 0.90 \times 65 \approx \text{EUR } 36.7/\text{t cement} \quad (2)$$

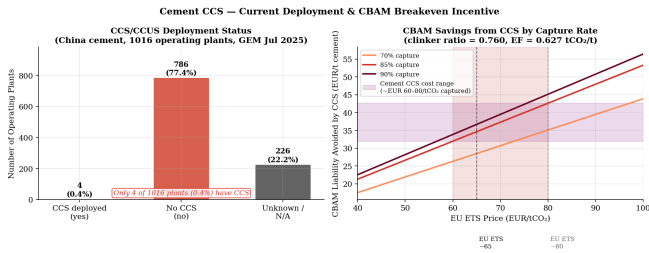


Fig. 7. Cement sector CCS deployment in China and CBAM-implied CCS breakeven cost. Of 1,016 GEM-tracked operating plants, only 4 report active CCS/CCUS (<0.4%). Breakeven assumes 90% CO₂ capture rate (conventional CCS captures ~85–90%; Homaei et al. 2026).

CCS becomes economically viable when this saving covers operating costs. At the prevailing CCS cost of EUR 60–80/tCO₂ captured (Al-Juaied and Whitmore, 2023) (= EUR 33.7–45.1/t cement), the breakeven EU ETS price is approximately EUR 65–80/tCO₂—placing current EU ETS levels at the lower bound of CCS viability for cement.

The technical floor for clinker content in ordinary Portland cement is 0.65–0.70. China’s primary SCMs—fly ash and blast furnace slag—are byproducts of sectors undergoing structural decline, constraining the clinker substitution pathway and reinforcing the long-run indispensability of CCS.

C. CN-ETS Reform: Before and After CBAM

China launched its national ETS in February 2021, initially covering only the power generation sector ($\approx 40\%$ of total CO₂ emissions) (MEE, 2021). The intensity-based allocation mechanism does not constrain absolute emissions, and allowance prices have traded in the range of CNY 40–105/tCO₂ (EUR 5–13/tCO₂)—far below EU ETS levels.

MEE has since announced three central reforms aligned with CBAM’s definitive phase: (i) transition from intensity-based to absolute emissions caps by 2027; (ii) sectoral expansion, with steel and cement incorporated into CN-ETS in 2024 and aluminium targeted for inclusion in 2025–2026; and (iii) introduction of allowance auctioning to replace free allocation. Attribution of these reforms solely to CBAM pressure requires caution: the dual carbon goals (*shuang tan*)—peak CO₂ by 2030, neutrality by 2060—provided independent domestic momentum. CBAM is best understood as correlating with—and potentially reinforcing—an existing trajectory, consistent with the analytical framing, though causal attribution is not asserted.

D. Political Economy Constraints

Three structural constraints govern the pace of CN-ETS reform and industrial decarbonization.

SOE dominance in steel and cement creates an asymmetric incentive structure: SOEs internalize compliance costs but do not fully capture the export competitiveness benefits of decarbonization, consistently favouring slower reform and more generous free allocation (Nahm and Urpelainen, 2021).

Energy security imperatives constrain the pace of transition away from coal-based steelmaking. Blast furnace operations are integrated with domestic coking coal supply chains

and concentrated in provinces—Hebei, Shanxi, Liaoning—where industrial employment is structurally dependent on BF-BOF production.

Regional heterogeneity in carbon intensity complicates uniform carbon pricing, generating distributional tensions that constrain the political feasibility of stringent absolute caps.

The two-level game framework (Putnam, 1988) is used here descriptively: the three constraints above define the boundaries of China’s domestic win-set, identifying what range of reform outcomes is politically feasible rather than predicting which outcome will occur. CBAM shifts the fiscal calculus of the status quo, but the framework does not predict whether that shift will translate into specific policy outcomes. Putnam’s original formulation was developed for democratic states with formalized legislative ratification; China’s win-set is shaped through party channels and SOE advisory bodies rather than legislative veto, which limits the framework’s direct applicability. It is retained as an analytical lens for mapping constraints, not as a predictive model. These constraints suggest reform trajectories will be gradual, sectoral, and subject to implementation delays, but causal attribution to CBAM alone is not asserted.

VII. DISCUSSION

The empirical findings in Sections V and VI establish three observations that anchor the discussion: per-tonne CBAM liability is dominated by the 21:1 BF-BOF/EAF liability ratio and China’s 17.6:1 fleet-average-to-EAF emissions-intensity ratio (Table II); CN-ETS pricing operates as a fiscal lever through Regulation EU 2023/956’s Article 9 deduction; and the political economy of reform (Section VI-D) is geographically concentrated, with five provinces bearing the bulk of exposure. We translate these into recommendations across three audiences—firms, policymakers, and the EU–China relationship—and close by naming a framework limit that conditions all three.

A. Enterprise-Level Strategies for Reducing CBAM Exposure

Section V-B establishes that CBAM liability is driven by per-unit carbon intensity. Firm-level action should therefore target the intensity gap directly rather than aggregate emissions reduction.

For steel exporters, the EUR 134.0/t per-tonne differential between BF-BOF and EAF at $P_{EU} = \text{EUR } 80/\text{tCO}_2$ defines the upper bound of within-firm CBAM-cost arbitrage. Three adjustment pathways follow. *First*, dedicated EAF capacity for EU-bound product lines exploits the BF-BOF/EAF intensity differential quantified in Section VI-A; coastal provinces (Jiangsu, Guangdong) with scrap-supply access are the natural candidates (Chen et al., 2023). *Second*, BF-BOF process adjustment—scrap charging and partial hydrogen injection—can reduce intensity meaningfully short of full technology replacement, providing a transition path during the 2025–2030 scrap constraint window. *Third*, mid-term portfolio rebalancing should treat CBAM exposure as a binding allocation parameter: EU-bound order books

should preferentially be filled from lower-intensity capacity, with high-intensity capacity redirected to domestic and non-CBAM export markets. The latter is not an emissions reduction strategy—global emissions are unchanged—but it is a CBAM-cost mitigation strategy entirely consistent with current Regulation EU 2023/956 scope.

For cement producers, decarbonization options are narrower. The thermodynamic irreducibility of calcination means that supplementary cementitious material substitution and CCS deployment are the only material pathways, and Eq. 2 establishes that CCS breakeven sits at the lower bound of current EU ETS prices. Cement firms should prioritize MRV readiness now: the operational distinction between EU ETS-aligned facility-specific emission factors and the punitive default values applied where MRV is absent is large enough to justify near-term investment in ISO 14064-aligned carbon accounting, independent of CCS deployment timing.

Across both sectors, a common firm-level imperative is documentation. Sectors not yet inside CN-ETS face the maximum CBAM liability under the current price gap (Table II, “not covered” row). Firms should build the data infrastructure now so that domestic payments are immediately offsettable once CN-ETS coverage extends—a sequencing question rather than a capability question.

B. Carbon Market Reform: Closing the Price Gap to Retain Revenue

Sections V-B and VI-A establish the revenue-retention logic: each euro increase in P_{CN} shifts an equivalent quantum of fiscal capacity from the EU budget to the Chinese budget, bounded above by P_{EU} . This reframes the carbon-pricing question from one of domestic distribution to one of international fiscal allocation, and changes the political economy of reform accordingly. Four reform vectors follow.

Absolute caps: the announced 2027 transition from intensity-based to absolute caps is the necessary condition for sustained price discovery. Table II shows that reaching $P_{CN} = \text{EUR } 40/\text{tCO}_2$ by 2030 would reduce per-tonne CBAM liability by 40% across BF-BOF, EAF, and cement at $P_{EU} = \text{EUR } 80$; reaching EUR 60 would reduce it by 70%. Cap stringency is the key calibration parameter.

Sectoral expansion: bringing aluminium into CN-ETS in 2025–2026, on the announced timeline, before CBAM aluminium scope deepens, captures the revenue-retention benefit at its largest. The sequencing matters: each year a sector remains outside CN-ETS while CBAM-liable is a year of pure fiscal outflow.

Auctioning and price floors: introducing a gradually rising auction share, with a reserve price calibrated to the threshold where CN-ETS payments begin materially offsetting CBAM costs, would convert the revenue-retention logic from a notional price signal into a realized fiscal instrument. Free allocation, by contrast, leaves the deduction clause working only at the margin.

Revenue recycling: the fiscal proceeds, once realized, should be earmarked for the technology and readiness gaps

identified in Sections VI-A–VI—EAF and scrap infrastructure subsidies, enterprise MRV grants. This closes the loop between the revenue-retention mechanism and the underlying decarbonization that CBAM is intended to drive.

The political economy constraints in Section VI-D condition all four vectors. SOE-dominated steel and cement firms in Hebei, Shanxi, and Liaoning bear the visible costs; the fiscal benefits accrue centrally. This intergovernmental asymmetry is a real constraint on cap stringency, and any reform path that ignores it will encounter implementation resistance regardless of central ambition.

C. EU–China Cooperation: From Coercive Pressure to Market Alignment

The carbon price gap (Section V-B) and the 30–60% offset ceiling (Li et al., 2026) define the fiscal stakes of the EU–China carbon-policy interaction. Cooperation can raise that ceiling through a phased pathway, but cooperation requires a framing shift that we want to name explicitly.

The dominant framing of CBAM and China treats the EU as agenda-setter and China as respondent: an external incentive structure produces an internal Chinese reform response. This framing is embedded in the analytical apparatus we use—Putnam’s two-level games framework presupposes external pressure transmitting into domestic politics—and we have followed that apparatus throughout the paper. But the framing is incomplete in a way that bears directly on cooperation prospects.

China’s dual-carbon goals (announced September 2020) predate CBAM’s adoption (May 2023). CN-ETS was launched (February 2021) before CBAM’s transitional phase began (October 2023). China is the core manufacturer of the EU’s green transition—photovoltaic modules, batteries, electric vehicles, wind turbine components—and the EU’s 2030 climate ambition is structurally dependent on Chinese industrial capacity. China is also a principal voice in the G77+China coalition, defending the principle of common-but-differentiated responsibilities (CBDR) and pressing for climate finance commitments from developed nations. A more accurate description of the empirical pattern in Section VI is therefore that CBAM *accelerates and sequences* a reform trajectory that China is already pursuing on its own terms; CBAM does not initiate that trajectory. This distinction is consequential for cooperation: a cooperation framework premised on China-as-respondent is unlikely to elicit the genuine mutual recognition that effective MRV equivalence requires.

With this framing in place, three cooperation pathways are available.

Short term (2025–27): negotiate mutual recognition of CN-ETS payments as CBAM offsets at facility level. The current 30–60% offset rate (Li et al., 2026) reflects in part the absence of formal MRV equivalence; a recognition agreement could lift the offset rate toward 80–90% conditional on third-party-verifier accreditation. This is the cooperation step with the largest immediate fiscal return for China and the lowest political cost for the EU.

Medium term (2027–30): harmonize MRV standards through joint emission-factor methodologies and mutual recognition of accredited verifiers, allowing Chinese installation-level data to replace CBAM default values at scale. This requires sustained technical dialogue between MEE and DG TAXUD, and is the operational precondition for any future linkage.

Long term (post-2030): if the price-gap convergence trajectory in Section V narrows the gap sufficiently, explore one-way market linkage on the model of EU–Switzerland ETS linkage. Linkage is technically demanding but politically efficient—it forecloses the entire CBAM-on-China question by aligning prices directly.

The CN-ETS–CBAM interaction also offers a transferable template. The revenue-retention logic gives other developing-country ETS jurisdictions—South Korea, Indonesia, Thailand, and potentially India—a fiscal rather than purely environmental rationale for domestic carbon pricing reform. To the extent this template diffuses, the significance of CBAM extends beyond the EU–China bilateral relation into broader carbon-policy architecture across Asia.

VIII. CONCLUSION

This paper has examined how the EU’s Carbon Border Adjustment Mechanism reshapes China’s carbon pricing architecture and sectoral decarbonization pathways. Drawing on plant-level data covering 458 steel and 1,016 cement facilities, and a CBAM liability model derived directly from Regulation EU 2023/956, we identified three transmission channels—a direct cost channel, a revenue retention incentive, and a signalling channel—through which CBAM’s external design produces internal pressure on Chinese policy.

Three findings carry the argument. *First*, per-tonne CBAM liability is dominated by technology heterogeneity rather than aggregate sector exposure: at $P_{EU} = \text{EUR } 80$, BF-BOF steel faces a net liability of EUR 140.7/t against EUR 6.7/t for EAF, a 21:1 per-tonne liability ratio that maps technology choice directly onto trade competitiveness (China’s fleet-average-to-EAF emissions-intensity ratio is 17.6:1). *Second*, the deduction clause in Regulation EU 2023/956 transforms domestic carbon pricing from a purely environmental instrument into a fiscal substitution mechanism: each euro of CN-ETS price retained in China is a euro not transferred to the EU budget. This reframing aligns the political economy of CN-ETS reform with treasury rather than environmental ministry incentives. *Third*, the empirical timing pattern of Chinese policy—steel and cement entering CN-ETS in 2024, the announced 2027 absolute-cap transition, sectoral expansion—is consistent with, but does not demonstrate, CBAM-induced reform acceleration. China’s dual-carbon goals provide independent momentum that predates CBAM. The most defensible reading is that CBAM functions as accelerator and sequencing constraint, not as initiating cause.

The policy implications follow from these findings. For Chinese firms, the per-unit intensity gap is the binding parameter; for the Chinese government, accelerating CN-ETS

price discovery through absolute caps and auctioning converts CBAM exposure into domestic fiscal capacity; for the EU–China relationship, mutual recognition of MRV equivalence offers the largest near-term gain at lowest political cost. The transferable element is the revenue-retention logic itself, which gives other developing-country ETS jurisdictions a fiscal rationale for carbon-pricing reform that environmental arguments alone have not generated.

The study has two principal limitations. Provincial cost allocations rely on capacity shares as a proxy for trade-flow shares, which may misattribute exposure where production specialisation diverges from export specialisation. Most consequentially, the correlation between CBAM implementation and CN-ETS reform acceleration is consistent with the analytical framework but does not constitute formal causal identification; we interpret the pattern as theoretically informed association rather than demonstrated effect. The two-level games framework we adopt presupposes external pressure transmitting into domestic policy, a presupposition that fits the mechanism we trace but understates the degree to which Chinese climate policy is shaped by domestic agenda-setting independent of external pressure.

Three directions for future research follow. First, customs-level microdata on EU-bound exports would allow disentangling production-mix from destination-mix effects that capacity-share proxies cannot resolve. Second, comparative analysis of CBAM-exposed jurisdictions with and without independent decarbonization trajectories would provide the closest available approximation to counterfactual identification. Third, a theoretical framework for non-Western agency in global climate governance, in which countries like China are treated as co-shapers rather than respondents, remains an open project that the CBAM–China interaction is particularly well-placed to inform.

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